June 16, 2025

Centers for Medicare & Medicaid Services
Department of Health and Human Services
Attention: CMS–0042–NC
P.O. Box 8013
Baltimore, MD 21244–8013

Subject: Comment on CMS–0042–NC—Request for Information on the Health Technology Ecosystem

To Whom It May Concern:

Oura welcomes the opportunity to provide comments in response to the Centers for Medicare & Medicaid Services (CMS) and the Office of the National Coordinator for Health Information Technology (ONC) Request for Information (RFI) titled "Health Technology Ecosystem" (CMS–0042–NC). We agree that the "[e]ffective and responsible adoption of technology can empower patients to make better decisions for their health and well-being" and have been building consumer-facing tools for over a decade—backed by science—to do just this.

We're facing an unprecedented array of health crises in our country today, with rising costs, faltering life expectancy, and an epidemic of chronic disease. We need innovative new approaches that can provide higher-quality care at lower cost.

Wearable technologies, especially Oura Ring, are uniquely positioned to mitigate these challenges—helping bridge the gap between clinical care and everyday health behavior. Wearables have already reached a high level of consumer acceptability: the 2024 Rock Health Consumer Adoption Survey found that ownership of wearable health devices in the United States ranges from 32% in the geriatric population to 66% of millennials. Beyond ownership and personal usage of these devices for fitness tracking at home, many wearable owners are also ready and willing to share this data with their healthcare providers. The combination of current market proliferation with the opportunity to expand access to devices—and willingness of users to share data with healthcare providers—supports the feasibility of bold new interventions at a national scale.

We appreciate CMS' and ONC's leadership in advancing a patient-centric, interoperable digital health ecosystem. Our comments focus on several areas outlined in the RFI that directly intersect with our technology and mission. We are ready and eager to partner with you on this important initiative.

Regards,

Ricky Bloomfield, MD
Chief Medical Officer, Oura

- **Patients and Caregivers**
    - PC–1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?
        i. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?
            - Improve my motivation to tackle challenging preventive and chronic health problems through relevant and timely action-oriented insights
            - Better understand my own health by viewing all of my health information in one place, including both health history as well as patient-generated (e.g., sensor) data
            - Help manage conditions for myself or my loved ones, including sharing relevant health data with them so they are informed
            - Share relevant health information with my care team, including sensor-based metrics or alerts from my wearable device
            - Schedule appointments with my care team
            - View and edit my care plan
            - Find accurate information about my costs
            - Find providers or other covered services in my health insurance network
        ii. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.
            - Help me improve my health literacy through accurate and timely educational material
            - Alert me when data patterns suggest early signs of illness or when recovery is impaired
            - Provide personalized insights about trends and contextualize them with medical data
            - Guide me through chronic condition management (e.g., optimizing metabolism & nutrition for pre-diabetes)
            - Connect me with my care team (including making appointments for me) when there is important action they need to take (e.g., write a prescription or order a test)
            - Engage with my insurer when something goes wrong to make sure my services are covered
    - PC–5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?
            - **Increase Awareness:** there are so many digital health products that it can be difficult to know which may be most effective. As noted below and in PC-7, CMS could create a registry of products

whose efficacy is supported by evidence of improved outcomes and positive behavior change.

- **Reduce Friction:** to reduce the burden of so many patient portals and improve consumer data access, CMS should mandate acceptance of third-party IAL2 credentials by EHRs, providers, and payers. Key challenges for consumer-facing apps include:
    1. **Endpoint discovery:** Finding FHIR endpoints is difficult; CMS/ONC should publish a centralized directory.
    2. **Onboarding:** Data holders are slow to register apps; standardized, mandated onboarding processes are needed.
    3. **Support:** Standards-based APIs receive inadequate support compared to proprietary systems, hindering access. Without better support, users often abandon these digital health tools. CMS and ONC must require stronger support of these APIs and easier authorization for apps acting on users' behalf.

i. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

- CMS should retain a light touch on app approval, but encourage attestation to codes of conduct (e.g., the CARIN Code of Conduct) that enable apps to declare a high bar for privacy and security. CMS could accept submissions of outcomes data associated with these apps, and publish the apps in a registry along with their outcomes data and code of conduct attestation (similar to the CARIN MyHealthApplication website).
- CMS should lower the barrier to access for proven digital health tools, including evidence-based wearable devices, through the following mechanisms:
    a. **Medicare Coverage Expansion:** cover wearable devices and apps under Medicare as durable medical equipment or remote monitoring tools.
    b. **Medicaid Waivers and State Innovation:** allow states to cover wearable devices via Medicaid 1115 waivers or incentives. The federal government could create a streamlined template for states to use.
    c. **Medicare Advantage and Private Plan Benefits:** encourage/require MA and private plans to offer

evidence-based wearable devices as wellness or chronic care benefits.

- Additional recommendations:
    a. ONC should change the definition of CEHRT (Certified Electronic Health Record technology) to “API CEHRT” (Application Programming Interface Certified Electronic Health Retrieval Technology).
    b. The ONC is able to redefine API CEHRT based on the original definition in the HITECH Act, which includes the adoption of standards in Section 3004 and defines health information technology as inclusive all of the “hardware, software, … or packaged solutions sold as services that are designed for or support the use by health care entities or patients for the electronic creation, maintenance, access, or exchange of health information.” This definition would include EHRs, LIS, RIS, payer systems, API vendors, and cloud-based solutions, in effect any HIT that supports some or all ePHI or EHI.
    c. API CEHRT should focus exclusively on certifying the HL7 FHIR APIs that providers and payers send and receive and no longer certify the functionality within the EHRs or EDI X12 transactions.
    d. CMS should then point to the new definition of API CEHRT in their regulations to ensure consistency between providers and payers
    e. The APIs on ONC’s Inferno Test Kit website should be supported by payers and providers and implemented within the next two years.
    f. Improve testing of APIs by requiring the inclusion of a standard test patient in the production instance of all HL7 FHIR-based API servers, and publish detailed results of these automated tests publicly on Lantern every week (with a score out of 100).

- PC–6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?
    - Digital health tools should follow platform-specific best practices for accessibility.
    - Digital health tools should follow best practices for user-centered design.
    - As noted above, some of the biggest challenges that consumers – and especially Medicare beneficiaries – experience revolve around the friction that is created when trying to access health information. Challenges include remembering usernames and passwords, disparate data sources and the challenges of tools connecting to those data sources, incomplete

data, and many more. As outlined above, we encourage CMS to consider ways to reduce the friction associated with connecting to providers and payers while increasing the utility of PHRs and other health navigation applications by making sure current regulations are followed by EHRs, providers, payer-technology vendors, and CMS-regulated insurers.

- PC–7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?
    - As noted above, CMS could create a registry of health applications and request submissions of outcomes data tied to these apps. This could be as simple as a link to PubMed or other journal articles. AI could be used to summarize key points in a user-friendly way, or app owners could submit their own summaries.
    - CMS should also offer funding to support outcomes-related research involving digital health tools. CMS may also be able to find this outcomes data in health tech partnerships and case studies that are written after treatment-control studies. CMS could put out a request for these.
- PC–8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?
    i. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?
        - Imaging data, including radiology images and pathology slides
        - After visit summary data, which is often not available via API access today
        - Claims data is available, but APIs are inconsistently implemented and difficult to access
    ii. b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?
        - See above
    iii. c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?
        - One opportunity for improving data integration is to support not only read access for patients but also structured, write-enabled interactions through trusted third-party applications. This could allow individuals to submit patient generated (e.g., home monitoring) health data, self-reported health updates, or preferences that inform care decisions. CMS could consider pathways to encourage bidirectional exchange as part of advancing consumer-facing API functionality and digital engagement.

- PC–10. How is the Trusted Exchange Framework and Common Agreement (TEFCA) currently helping to advance patient access to health information in the real world?
    - TEFCA is moving closer toward becoming an operational, everyday system for health information exchange. TEFCA has established the necessary framework and governance structures (QHINs, Participants, and Subparticipants) that will enable nationwide health information exchange, including Individual Access Services (IAS).
    - The implementation of the 21st Century Cures Act, alongside TEFCA development, has pushed policy toward patient ownership of data. IAS within TEFCA will enable individuals, through the application of their choice, to query and obtain their medical records and other health information from multiple sources, including the ability to discover where their records may reside through record location technology.

  i. What changes would you suggest?
    - To advance TEFCA implementation for patient access, we recommend:
        a. Implement standardized identity proofing using IAL2 credentialing with NIST 800-63-3 standards and the CARIN Credential Policy for trust alignment, with federation via UDAP Trust Community.
        b. Set a nearer-term timeline by which all Designated QHINs must complete cross-network IAS testing on both sides of document-based exchange (that is, as a requesting and responding QHIN).
        c. Update the RCE's RLS Directory with information indicating which OIDs are responding to IAS Queries, and which have successfully responded with records for IAS purposes, and make this information query-able. Over time, we would also expect this directory to interoperate with a national digital healthcare directory, which we hope would represent the parent-child relationships of different OIDs.
        d. Modify its governance structure to ensure more representation by IASPs, specifically on the Governing Council, and the Participant/Subparticipant Caucus.
        e. Clarify that TEFCA's reciprocity requirements are different for non-HIPAA IASPs than for other Participants and Sub-Participants. OCR has long-standing guidance that patient requests cannot be conditioned on acceptance of other obligations. Also, the TEFCA governing documents support a pathway for IASPs to make a choice whether to support reciprocal exchange, but that must be with the patient's consent. The TEFCA Conditions of Participation

makes clear that IASPs that are Non-HIPAA Entities are not required to respond, but given the stance that several QHINs are taking, we believe a crystal clear policy clarification is needed, to remove conditions of reciprocity as a current barrier to wider IAS exchange on TEFCA.

f. Address advancing privacy & consent management capabilities using identity-linked consent artifacts, OAuth2-based user-managed authorization flows, and federated consent via UDAP Authorization Extension Objects, while using TEFCA as a lever to raise the necessary capabilities of the TEFCA participants/ subparticipants to adopt essential privacy & consent management capabilities ensuring that data is only shared with those who are authorized to receive that data based on applicable privacy rules and patient data sharing consent directives. Examples of the latter may be collaboration with HL7 FAST Consent Management and HL7 Sensitive Data projects towards inclusion into TEFCA at the right time.
g. Simplify technical integration through SMART on FHIR for initial access, UDAP Dynamic Client Registration for scalable onboarding, and certificate-based client identity with digital certificate validation.
h. Reduce privacy and security compliance burdens by adopting HL7 FAST Security IG, implementing UDAP JWT-based client authentication, and using privacy-protective attribute release.
i. Improve patient trust through reusable IAL2 credentials from certified CSPs, implementing the CARIN Credential Policy for policy equivalency, and supporting multiple authentication pathways.
j. Establish credential federation, allowing digital identities verified by one Credential Service Provider to be accepted by another without requiring duplicate verification. The HL7 FAST Security and Identity IGs provide stakeholder consensus-based approaches to this.

ii. What use cases could have a significant impact if implemented through TEFCA?

- High-impact use cases for TEFCA implementation include:
    a. IAL2 credential with Record Locator Services using SMART on FHIR EHR portal authentication, enabling consumers to access health information across multiple provider portals with a single credential.

b. IAL2 credential with XCPD demographics match across multiple Health Information Exchanges, creating a single credential usable across multiple health information networks.
c. IAL2 credential using OpenID Connect with single or multiple QHINs, returning either CDA or FHIR resources to IAS Providers, streamlining authentication and supporting both document and API-based access.
d. Facilitated FHIR using HL7 FAST Security Implementation Guide, allowing QHINs to return FHIR endpoints instead of data directly for secure querying.
e. Facilitated FHIR with B2B User Authorization Extension Object, supporting complex authorization models across both consumer and business use cases.
f. Cross-provider credential acceptance demonstrating one credential issued by one CSP could be accepted by another CSP, establishing true identity federation in healthcare.

TEFCA could unlock new value by supporting direct patient contributions to their health record using a combination of structured FHIR write transactions and other means for clinicians to access patient contributions, e.g., access to their PHR app/portal where direct writes are not appropriate. For example, enabling a patient to upload medication adherence logs or self-reported health goals through an application of their choice – validated by a credentialed identity – could significantly enhance data completeness and enable more personalized care planning, especially for chronic disease management.

iii. What standards are you aware of that are currently working well to advance access and existing exchange purposes?
- SMART on FHIR is widely implemented across the industry and provides immediate feasibility for IAS exchange
- OAuth 2.0 technologies effectively support IAL2 credentials aligned with NIST requirements
- The IHE Document Exchange XCPD API support secure, interoperable access for demographic matching between systems

iv. What standards are you aware of that are not currently in wide use, but could improve data access and integration?
- HL7 FAST Security (required standard by January 1, 2026, under TEFCA) offers improved capabilities including certificate-based authentication and enhanced security controls

        - HL7 FAST Identity includes requirements for creating, managing, and federating Digital Identity including a Digital Identifier, as well as best practice patient matching, leveraging the certificate-based trust in HL7 FAST Security
        - OpenID Connect Federation 1.0 enables credential federation across different service providers
        - NIST 800-63-3 IAL2 compliant credentials, PKI certificate validation, and JWT with federation extensions strengthen authentication
        - The NCPDP Patient Experience Identifier improves patient matching without compromising privacy
        - UDAP Dynamic Client Registration enables scalable onboarding of applications
    v. Are there redundant standards, protocols, or channels that should be consolidated?
        - Fragmented identity verification standards across QHINs, Participants, and Subparticipants create redundant verification processes
        - Non-standardized acceptance of IAL2 credentials creates unnecessary friction for consumers accessing health information across multiple systems
        - The TEFCA technical framework needs standardized methods for passing authenticated credentials between QHINs. Toward this end, accrediting bodies should publish the endpoint for keys used exclusively to sign services that meet accreditation requirements, along with plain language stating the minimum identity and authentication assurance for which a key at that endpoint can be relied upon, as a helpful step to increase confidence in the binding with the Identity Provider.
        - Different trust frameworks (Carequality, DirectTrust) need a common credentialing policy establishing equivalency through standardized requirements for IAL2 and PKI credentials
    vi. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?
        - Direct connections via standard FHIR APIs have worked well to bootstrap the industry; however, it's still complex for digital health tools to connect to thousands of disparate endpoints. Thus, the need for a single endpoint that enables patients to access all of their data in a standard and reliable way is pressing.
- PC–12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?
    - CMS should work to enable users to share their patient-generated health data with their care team in standard ways (see feedback above about

standard ways to write data back to the EHR and/or access to the patient's data on their app/portal of choice). This may include information such as:
    - Activity
    - Sleep quality
    - Overnight SpO2
    - Resting heart rate
    - Lab results from external sources
    - Links to images
    - Alerts about potential illness (or other regulated features)
  - Digital presentation of insurance coverage during appointment scheduling: a standardized, QR-enabled insurance card format would allow for instantaneous eligibility confirmation, helping patients avoid surprises related to coverage and network status. CMS could support this by naming a national digital insurance card standard and encouraging health plans to implement it via their Patient Access API infrastructure.

- **Providers**
  - PR–1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC–5) digital health products for their patients?
    - **Promote innovative pilot opportunities:** enable small pilots across ACOs that can quickly determine efficacy of wearable devices in a variety of clinical scenarios. Use these results to scale the interventions that are shown to be most effective.
    - **Incentivize integration:** Provide targeted reimbursement models or grants that support the integration of approved digital tools into clinical workflows, especially for chronic condition monitoring, sleep health, and preventive care.
    - **Expand technical assistance:** Offer implementation support and education for rural providers to reduce barriers to adoption, including infrastructure limitations and technical unfamiliarity.
    - **Support outcome-linked approval pathways:** Establish a transparent, streamlined approval framework where digital health products demonstrating real-world efficacy in engagement, adherence, or clinical outcomes are recognized and promoted for provider use.

    i. What are the current obstacles?
      - **Reimbursement uncertainty:** Lack of clear and consistent reimbursement mechanisms for patient-generated health data (PGHD) and remote monitoring limits provider incentive.
      - **Technical and workflow integration challenges:** Many providers – especially those in resource-constrained settings – lack the EHR interoperability or infrastructure to integrate data from digital tools like wearables.

        - **Trust and liability concerns:** Providers may hesitate to recommend tools without FDA clearance or clearly defined clinical guidelines on how to use the data effectively.
    ii. What information should providers share with patients when using digital products in the provision of their care?
        - Digital health tools have the potential to augment traditional care in lower-cost ways by providing actionable insights. So it's important to clearly explain the purpose and benefits of the digital product (e.g., sleep tracking for identifying patterns related to metabolic or mental health).
        - Outline how the data will be used in decision-making or care planning
        - Share any data privacy considerations, including who can access the data and how it is stored
        - Set expectations about patient responsibilities (e.g., consistent use, syncing data) and what kind of feedback or actions may result from the data
    iii. What responsibilities do providers have when recommending use of a digital product by a patient?
        - Recommending products that are evidence-based, approved (if applicable), and relevant to the patient's clinical context
        - Ensuring patients are educated on proper use, including limitations of the technology
        - Integrating insights responsibly into care, ensuring that PGHD is contextualized alongside clinical data and does not replace clinical judgment
        - Protecting patient privacy and complying with HIPAA when handling data transmitted from consumer-grade devices
- PR–4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?
    - Administrative workflows should accommodate not just viewing of provider schedules or forms but also patient-generated inputs – for example, enabling patients to pre-populate intake forms or submit supplemental data like medication logs or social history directly through apps. CMS could clarify that certified APIs should include FHIR write support for this class of interactions, supported by patient consent and identity validation mechanisms.
    - One overlooked opportunity is the modernization of eligibility verification and benefits display using standards-based digital insurance cards. CMS should support the use of open, API-based insurance card formats – such as the CARIN Digital Insurance Card IG – which provide real-time coverage details via a QR code or app display. This allows patients to

carry accurate, up-to-date coverage information, streamlining registration, scheduling, and intake workflows without relying on legacy manual or paper-based processes.

- CMS should clarify that administrative workflows, such as prior authorization, should support **patient-directed data access**. This includes surfacing PA status updates through consumer-facing APIs that are connected to the same adjudication engines used in provider-facing workflows. Aligning provider and patient experiences reduces confusion, improves adherence, and reinforces the role of the consumer as an informed participant in the care journey.

- PR–9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?
    - CMS should require that any system used for registration and login to access data on any health care network follow the NIST 800-63-3 digital identity guidelines for identity assurance level 2 (IAL2) and authenticator assurance level 2 (AAL2) for patient and provider identity and authentication. This requirement could be implemented within certification programs for providers and payers.
    - CMS can accelerate adoption of trusted identity credentials by endorsing federated digital identity models that allow patients to use one IAL2-verified login (e.g., via CLEAR, ID.me, Login.gov) across systems. This approach was successfully demonstrated by CARIN in partnership with CMS, CommonWell, and athenahealth. Providers will need implementation support and clear certification expectations to ensure that identity standards are met uniformly across networks.

  i. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?
        - Requiring the NIST 800-63-3 digital identity guidelines for identity assurance level 2 (IAL2) and authenticator assurance level 2 (AAL2) for patient and provider identity and authentication would strengthen identity and access management capabilities for each individual user on any network (providers, health plans, and individuals). This balances patient privacy by building transparency and trust that individuals are who they say they are and affords better identification of who each individual represents on any network.
        - Endorsing and encouraging compliance with industry standard privacy frameworks, such as the NIST Privacy Framework to provide a clear path for digital health companies to develop compliant tools and services that patients can both trust and access conveniently.

- **Payers**
  - PA–2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?
    i. See our response to PC-5 above. One of the most powerful ways to encourage compliance with current requirements is to transparently publish compliance, similar to what Flexpa has done: https://www.flexpa.com/blog/may-2025-state-of-the-payer-patient-access-api-report
- **Technology Vendors, Data Providers, and Networks**
  - TD–1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?
    i. **Launch a CMS-approved digital product directory or sandbox:** Establish a centralized registry of approved or verified third-party applications, including those focused on preventive care, sleep, and chronic condition management. A testing environment would lower the barrier for new entrants and validate interoperability.
    ii. **Offer clear pathways for non-coverage product validation:** Provide guidance on how digital health products can be evaluated based on real-world performance, health outcomes, and user engagement without requiring a full coverage determination. This encourages innovation while maintaining guardrails.
    iii. **Explore offering reimbursement for evidence-based digital health tools:** tools such as wearables that demonstrate improved outcomes are often less expensive than comparable “clinical-grade“ medical devices.
  - TD–5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?
    i. Having a central place for digital health tool developers to access FHIR endpoints is a good start, although not sufficient if the implementation of these endpoints is inconsistent or it’s challenging to connect because of significant friction for app vendors to register. Ultimately, a central TEFCA via IAS would solve this problem, although in the spirit of optionality to see where the market goes, supporting a bootstrapped version of a directory would be a great first step. This could be hosted on Lantern with minimal additional cost.